

Mathematics for Machine Learning: Homework 6

1. Calculate the integral

a) $\int_0^{2\pi} \cos^2 2x dx,$

c) $\int_0^{\ln 2} \frac{dx}{\sqrt{1+e^x}},$

b) $\int_0^1 \frac{x^2}{1+x^6} dx,$

d) $\int_{\frac{1}{e}}^e |\ln x| dx.$

2. Calculate the improper integral

a) $\int_0^1 \ln x dx,$

c) $\int_2^{+\infty} \frac{dx}{x^2+x-2},$

b) $\int_0^{+\infty} x 2^{-x} dx,$

d) $\int_0^{+\infty} \frac{x \ln x}{(1+x^2)^2} dx.$

3. Study the convergence of the improper integral

a) $\int_1^{+\infty} \frac{dx}{x \sqrt[3]{x^2+1}},$

c) $\int_0^2 \frac{\ln(1+\sqrt[5]{x^3})}{e^{\sin x}-1} dx,$

b) $\int_1^{+\infty} \frac{\operatorname{tg} \frac{1}{x}}{1+x\sqrt{x}} dx,$

d) $\int_0^1 \frac{e^x}{\sqrt{1-x^3}} dx.$

4. Find the Taylor series representation of the function f at the point $x_0 = 0$.

a) $f(x) = \cos^2 x,$

b) $f(x) = \frac{x}{1+x-2x^2}.$

5. Calculate the gradient of f .

a) $f(x, y) = x \sin(x + y),$

c) $f(x, y, z) = xyz e^{x+y+z},$

b) $f(x, y) = \operatorname{arctg} \frac{x+y}{1-xy},$

d) $f(x, y, z) = x^{\frac{y}{z}}.$

6. Find $\frac{\partial u}{\partial t}$ and $\frac{\partial u}{\partial s}$, if

a) $u = f(x), x = t^2 + s^2,$

b) $u = f(x, y), x = \sin t, y = \cos s,$

c) $u = f(x)g(y), x = ts, y = t - s,$

d) $u = f(x, y, z), x = ts, y = 2t + s, z = t - 3s.$